Clinical Risk Assessment Report: Patient 29

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 11.6% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.52x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which reduced the patient's absolute risk by 1.7%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.66 [Top 24%] (+1.2%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.6 [Bottom 18%] (+1.1%)**

Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

3. Protective / Mitigating Factors (Reducing Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.01 [Avg (74th %ile)] (-1.7%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Patient Age (Age) **Value: 66.0 [Avg (59th %ile)] (-1.6%)**

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.64 [Bottom 19%] (-1.3%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: 1.41 [Top 6%] (-1.2%)**

Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: -2.0 [Bottom 5%] (-1.1%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.01 [Top 21%] (-0.8%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Machine Learning Feature Attribution (SHAP Decision Path)